



Deep Learning Applications in Hierarchical Time Series Forecasting of Market Value

John Kamwele Mutinda¹ · Li Yong¹

Accepted: 26 May 2025

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

Accurate market value forecasting is essential for investors, financial analysts, and policymakers due to the inherent volatility and complexity of financial markets. This study employs hierarchical time series (HTS) forecasting to predict the market value of the DJIA index across three levels: overall market value, sector-level market value, and individual stock market value. To capture sequential dependencies in hierarchical financial data, deep learning neural networks—including recurrent neural networks (RNN), long short-term memory (LSTM), and gated recurrent units (GRU)—were utilized for base forecasts. To ensure forecast coherence, various reconciliation methods were applied, including Bottom-Up (BU), Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols). The effectiveness of these reconciliation approaches was evaluated using performance metrics such as average root mean squared error (AvgRMSE), average mean absolute percentage error (AvgMAPE), average relative RMSE (AvgRelRMSE), and average relative MAPE (AveRelMAPE). The findings indicate that reconciliation methods significantly improve forecast accuracy. Among them, MinT (WLS-struct) consistently achieved the lowest metrics across multiple forecast horizons, including short-term (115 days) and long-term (230 days) predictions for all deep learning models.

Keywords HTS · DJIA · RNN · LSTM · GRU · Forecast coherence · MinT (Wls-struct)

✉ John Kamwele Mutinda
john.mutinda@mail.ustc.edu.cn

Li Yong
yonglee@ustc.edu.cn

¹ School of Management, University of Science and Technology of China, Jinzhai Road, 230026 Hefei, People's Republic of China

1 Introduction

A time series is a collection of data points observed at regular intervals over time. Time series analysis is the process of examining the patterns, trends, and behaviors that emerge from data over time. This analysis can help explain previous behavior, anticipate future values, and establish causal relationship (Anderson, 2011). Time series analysis is widely utilized in various areas, including finance and economics (Chan, 2004; Tsay, 2005; Cochrane, 1997; Mudelsee, 2019, 2010; Thomson, 1990).

Hyndman and Athanasopoulos (2018), Wickramasuriya et al. (2018) define hierarchical time series as an arrangement of time series that can be aggregated at different levels. Hierarchical time series (HTS) forecasting has been an important field of study, especially in fields where data organization is hierarchical, such as retail sales (Villegas & Pedregal, 2018), solar energy (Yagli et al., 2019), tourism demand (Athanasopoulos et al., 2009), mortality rates (Shang & Hyndman, 2017), prison populations (Zheng et al., 2006), and wind power generation (Ye et al., 2019).

In this paper, we are skewed to the cross-sectional coherence and consider the HTS of market value. When determining market value, the market value of various stocks is combined to get the market value of industries, which is then added up to determine the total market value of the entire market index. It is logical upon forecasting that the overall market value represented at higher levels of the hierarchy should equal the sum of the values at lower levels. However, it is often difficult to comprehend and predict hierarchical time series data as forecasting techniques may find it difficult to grasp the relationships and interactions across many levels of aggregation (Nenova & May, 2016). This is due to the fact that forecasts are frequently made independently for every series, which results in discrepancies. These separate forecasts are reconciled using several Hierarchical Forecasting techniques to ensure coherence (Spiliotis et al., 2019).

Creating coherent forecasts is a difficult task for verifying the decision-making process at various levels of the hierarchy. Forecasts of individual series are coherent when they total up properly at all levels while maintaining the hierarchical structure. Coherence is required at either the cross-sectional or temporal levels. At the cross-sectional level, the shop-level demand for a specific product may be pooled at the regional or national levels. Forecasts at the day level must be consistent at the weekly level, then at the monthly level, and so on. The approaches to coherence change depending on the objectives and type of the data employed. This makes forecasting more difficult since forecasters now need to guarantee coherence in addition to selecting the optimal model for every product. It is essential to Hierarchical Forecasting because it guarantees that judgments at the same time at various levels.

Analytical methods for reconciling forecasts have been tabled by researchers. These methods not only facilitate the coherence of forecasts, they also lead to improvement of forecast accuracy. For example, the Bottom-up (BU) approach which is a primary HF method in which just the lowest level of the hierarchy is forecasted, and then the forecasts reconciled to project higher-level series (Dangerfield & Morris, 1992; Fliedner, 1999; Karmy & Maldonado, 2019; Widiarta et al., 2009). The top-down (TD) forecasting approach uses proportions to estimate lower-level series and then the BU is used to reconcile the bottom level forecasts (Gross & Sohl, 1990;

Fliedner, 1999; Karim & Maldonado, 2019; Hyndman et al., 2011; Widiarta et al., 2009). Middle-out (MO) combines BU and TD, making forecasts for a middle level and then aggregating or disaggregating them to anticipate higher and lower levels, respectively (Abolghasemi et al., 2022a). Linear combination is a recently proposed and now enhanced approach. The method essentially aggregates the individual forecasts provided for each node in the hierarchy in a manner that is linear, as the name implies. The strategy, known as trace minimization (MinT), minimizes the variance of prediction mistakes discovered on the trace of the variance-covariance matrix, hence lowering overall forecast variance. In a number of scenarios and forecasting challenges, MinT has proved to be a very efficient model. Mancuso et al. (2021), Abolghasemi et al. (2022b)

The advent of artificial intelligence (AI) has revolutionized numerous fields, particularly in time series forecasting and market predictions. As data volumes and complexity continue to grow, traditional statistical methods often struggle to capture intricate patterns in sequential data. Deep learning models like Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Recurrent Neural Networks (RNN) (Zheng et al., 2019; Pirani et al., 2022) have emerged as powerful tools to address these challenges. These models are designed to learn and retain long-term dependencies in sequential data, making them highly effective for forecasting tasks. LSTMs, with their ability to remember long-term patterns, excel in time series prediction, while GRUs offer a more efficient alternative with fewer parameters. RNNs, though simpler, form the backbone of many deep learning applications, providing a foundation for sequential data processing. Together, these models are reshaping AI's role in industries ranging from finance to healthcare, enabling more accurate, real-time predictions and decision-making.

Given the volatility of financial markets, investors, financial analysts, and lawmakers must examine market values at multiple levels of granularity. Forecasting market value at three levels—total aggregate market value, industry market value, and individual stock market value—is critical for investors because it gives an extensive understanding of market dynamics and potential investment opportunities. At the highest level, displaying the total aggregate market value of an index such as the DJIA assists investors in determining the overall market trend, allowing them to make informed decisions about whether to enter or quit the market. This comprehensive view is critical for portfolio managers and institutional investors who must connect their strategy with macroeconomic indicators and market sentiment. By anticipating market shifts, they can optimize their portfolios to mitigate risks and capitalize on growth opportunities, thereby enhancing their investment returns.

At the industry level, forecasting market values is particularly important for sector-specific investment strategies. Different industries respond uniquely to economic changes, technological advancements, and regulatory developments. Investors focusing on sector rotation strategies can use industry-level forecasts to identify which sectors are likely to outperform or underperform in the near future. This allows for strategic reallocation of assets to maximize returns. Finally, at the individual stock level, accurate forecasts enable investors to select specific stocks that are expected to appreciate in value. This granular analysis is vital for stock-picking strategies, where the goal is to identify undervalued stocks with strong growth potential. By

understanding the anticipated performance of individual stocks within the context of their respective industries and the broader market, investors can make more precise and profitable investment choices.

In this regard, this study seeks to analyze and compare the performance of several forecasting methods in the field of HTS analysis. The study's main objective is to deploy and compare deep learning neural networks, specifically RNN, LSTM, and GRU, in forecasting hierarchical time series data of market values while maintaining coherence across multiple forecasting horizons and aggregation levels.

The methodology of this study involves forecasting 37 series, consisting of one aggregate series, six industry-level series, and 30 individual stock series. Forecasting is conducted individually for each series using RNN, LSTM, and GRU models. Once individual forecasts are generated, they are reconciled using selected hierarchical forecast reconciliation methods. These methods include Bottom-Up (BU), Top-Down using forecast proportions (TD-Fp), Middle-Out with forecast proportions (MO-Fp), MinTrace with ordinary least squares (MT-Ols), MinTrace with weighted least squares (MT-Wls), and Optimal Combination using ordinary least squares (OC-Ols). The reconciliation ensures that forecasts across different levels of the hierarchy remain coherent, improving the reliability of the predictions for investment decisions.

To evaluate the performance of the forecasting models and reconciliation methods, several key metrics are computed, including Average Root Mean Squared Error (AvgRMSE), Average Mean Absolute Percentage Error (AvgMAPE), Average Relative Root Mean Squared Error (AvgRelRMSE), and Average Relative Mean Absolute Percentage Error (AvgRelMAPE). These metrics provide a comprehensive assessment of forecasting accuracy across different levels of the hierarchy. The results highlight the comparative effectiveness of different reconciliation methods, with MinTrace (Wls-struct) demonstrating consistent hierarchical forecasting accuracy across various levels, methods, and reconciliation schemes.

The contributions of this study are:

- This research, for the first time, prepared the market value of different stock companies, industries, and the entire composite index in a hierarchical structure.
- Deep learning neural networks were used in forecasting hierarchical time series data, generating forecasts that are coherent using different forecast reconciliation methods, thus setting the stage for deploying deep neural networks in time series that exhibit different granular levels.
- Forecasting and reconciliation were conducted using two different forecast horizons: half a trading year and one trading year, asserting the robustness of the proposed methodology.
- The study demonstrated a competitive performance of different forecasting methods and reconciliation techniques, with MinT (Wls-struct) showing a consistently high hierarchical forecasting accuracy across various levels, methods, and reconciliation schemes.

The manuscript is structured as follows: Section 2 identifies relevant work and research gaps. Section 3 describes the technique, which includes data preparation, model building, implementation details, and assessment measures. Section 4 summarizes the forecast outcomes. Finally, Section 5 ends the manuscript, and Section 6 discusses the study's limitations and recommendations for further research.

2 Related Work

Supply chain management, healthcare, economics, finance, and other sectors all benefit from the use of hierarchical time series forecasting. It's comparable to a combination of many techniques used to ensure forecasts make sense. First, we forecast using sophisticated techniques. Next, we employ several methods to modify these forecasts such that they substantially match each other (Mancuso et al., 2021; Hyndman et al., 2011; Mircetic et al., 2022; Gleason, 2020). Forecasting can be challenging due to the various levels at which data is arranged. Both large-scale and small-scale judgments must be made. Recently, scientists have started experimenting with several approaches to deal with issue and improve forecasts. Several studies have looked into different methods for forecasting time series data, such as neural networks, machine learning, and classical statistics.

For instance, Mahkya et al. (2017) conducted a study with the goal of improving the accuracy of Central Java export value forecasts, which have an important effect on the nation's economy. They employed Radial Basis Function Neural Networks (RBFNN), Hybrid ARIMA-RBFNNs, and Autoregressive Integrated Moving Averages (ARIMA) to forecast individual series at all hierarchy levels utilizing the Bottom-up technique. The outcomes demonstrated that for long-term forecasts, the bottom-up strategy works best.

Similarly, Silveira Gontijo and Azevedo Costa (2020) evaluated hourly electrical generation data in Brazil from 2018-2020. The data was grouped by electrical subsystems and energy sources, and the forecast accuracy of the ARIMA and Error, Trend, Seasonal (ETS) models was evaluated using three hierarchical methods: bottom-up (BU), top-down (TD), and optimal reconciliation. Their findings showed that optimal reconciliation models consistently outperformed the others throughout numerous prediction periods.

Abolghasemi et al. (2022b) highlighted the significance of hierarchical forecasting in supply chain decision-making. They examined a number of frequently employed hierarchical forecasting techniques, including optimum linear combination, bottom-up, and top-down methods. Additionally, they emphasized linear combination techniques, which frequently outperform conventional techniques by combining individual sales projections from all series and levels. Nevertheless, these traditional approaches might not take into account aspects like cost and prestige for promotions at various levels of the hierarchy, which could have an impact on how well they work.

In order to ensure coherence with aggregated predictions, Wickramasuriya et al. (2018) addressed the problem of forecast reconciliation in large collections of time series by making sure that forecasts for disaggregated series abide by aggregation restrictions. A different study by Hyndman et al. (2011) suggested a bottom-up, linear

regression strategy for optimal base forecasts. This approach is applicable to grouped and hierarchical time series and has demonstrated efficacy in forecasting monthly inflation and tourism demand.

Machine learning models, especially those based on neural networks, have augmented statistical methodologies for individual time series forecasting in academic literature. Novel machine learning algorithms have been presented in numerous studies (Yamak et al., 2019; Chen et al., 2021; Liu et al., 2020; Bontempi, 2013; Khan et al., 2024; Shahzad et al., 2024) to improve the accuracy of time series forecasts. For hierarchical time series forecasting, these machine learning techniques have also been modified. For example, a work by Spiliotis et al. (2021) suggests a bottom-up method to forecast bottom-level series: base forecasts for all series in the hierarchy are fed into machine learning models (such Random Forest and XGBoost). Next, the forecasts at the lowest level are combined to complete the reconciliation.

Several empirical studies have shown that bottom-up forecasting methods outperform top-down approaches, particularly in finance. Kinney (1971) found that disaggregating earnings data by market segments yielded more accurate predictions than depending just on firm-level data. Similarly, Box et al. (2015) studied telephone demand and found that lower-level forecasts were more accurate than top-down approaches. Furthermore, Ahmadi et al. (2020) studied yearly GDP growth rates across 18 nations and discovered that disaggregation boosted forecast accuracy significantly. These findings show the effectiveness of hierarchical time series techniques in financial forecasting, emphasizing the significance of merging particular company unit insights to improve prediction accuracy.

Time series forecasting has made great progress in recent years as a result of the increased usage of deep learning techniques. ANNs outperform standard statistical approaches in time series analysis (Chengzhao et al., 2019; Loureiro et al., 2018; Wang & Wang, 2015; Xu & Cohen, 2018; Asif et al., 2024). RNNs, a deep learning algorithm, improve time series data processing by maintaining information across multiple inputs (Zheng et al., 2019; Pirani et al., 2022). RNNs connect hidden layer units to better capture temporal relationships in data (Mandic & Chambers, 1999). Their sequential nature is useful for time series forecasting (Deng et al., 2018). RNNs face issues with vanishing gradients over extended time spans, which are addressed by LSTM networks (Hochreiter & Schmidhuber, 1997b). LSTM has been widely used to anticipate time series, yielding encouraging results. Wang et al. (2019), Karevan and Suykens (2020), Hochreiter and Schmidhuber (1997a), Wan et al. (2020), Bala et al. (2019). Despite growing interest in neural networks for time series analysis, their use in hierarchical time series analysis is limited.

While there is research on hierarchical time series analysis in a variety of domains, including finance and economics, there is a notable lack of comprehensive studies focusing on forecasting market values using hierarchical time series reconciliation methods and a variety of forecasting methods, including RNN and its variants such as LSTM and GRU models. This study aims to bridge this gap by comparing the performance of several forecasting algorithms inside a hierarchical time series framework. The purpose of this study is to provide major insights into the domains of hierarchical time series analysis and financial forecasting approaches by assessing the efficiency of several methods for estimating market values for the DJIA index.

2.1 Research Gap

There is still a substantial research gap in the application of HTS forecasting to financial market values, despite the substantial gains in HTS forecasting across other domains. The use of HTS in areas including energy production, supply chain management, and macroeconomic indicators has been thoroughly studied in previous research; however, its application to market indices has not received as much attention. Furthermore, although RNN and its variants, like LSTM and GRU, have demonstrated encouraging outcomes in time series forecasting, their potential within the HTS framework has not been thoroughly explored, especially with regard to financial market forecasting. This gap presents a critical opportunity to enhance the predictive accuracy and robustness of market value forecasts by leveraging advanced neural network architectures.

Ensuring coherence throughout the hierarchy's levels is a difficulty in HTS forecasting, as incoherent forecasts can add biases and lower overall accuracy. In order to maintain coherence throughout the hierarchical structure, this research attempts to address this problem by reconciling forecasts utilizing a variety of reconciliation techniques.

3 Data and Methods

3.1 Data

This study's analysis is based on a dataset that includes the market values of thirty companies from the Dow Jones Industrial Average (DJIA) market index, each of which represents a key industry participant. The daily market values for each company was calculated by multiplying the outstanding shares by the daily closing prices from January 4, 2021, to May 31, 2024. This time frame was chosen to provide a thorough overview of market dynamics in recent years, including key market events and trends. This data was sourced from Yahoo Finance Website.

3.2 Market Value Calculation

The market value of a company i on day t is calculated as:

$$\text{Market Value}_{i,t} = P_{i,t} \times \text{Shares Outstanding}_i$$

where:

- $\text{Market Value}_{i,t}$: The market value of company i on day t , in US dollars (USD).
- $P_{i,t}$: The closing stock price of company i on day t , in US dollars (USD).
- $\text{Shares Outstanding}_i$: The total number of shares outstanding for company i , which is a unit-less quantity as it is simply a count of shares.

To calculate the market value of an entire industry for a given day, we sum the market values of all companies in that industry:

$$\text{Market Value of Industry}_t = \sum_{i \in \text{Industry}} \text{Market Value}_{i,t}$$

where:

- **Market Value of Industry_t**: The total market value of all companies within a given industry on day t , in US dollars (USD).
- $\sum_{i \in \text{Industry}}$: Summation over all companies i in the given industry.

Finally, to calculate the aggregate market value across all industries for a given day, we sum the market values of all industries:

$$\text{Aggregate Market Value}_t = \sum_{\text{Industry}} \text{Market Value of Industry}_t$$

where:

- **Aggregate Market Value_t**: The total market value across all industries on day t , in US dollars (USD).
- $\sum_{\text{Industry}}$: Summation over all industries.

To facilitate our analysis, we aggregated the individual company data into six distinct industries, each representing a key segment of the market. These industries were further aggregated into total aggregate market value:

- **Level 0**: Representing the total market value of all companies combined.
- **Level 1**: Comprising industry-level aggregation, capturing the collective market value of companies within each industry.
- **Level 2**: Consisting of individual company data, providing detailed insights at the company level.

Furthermore, to make the research easier, organizations from less defined industrial groups were grouped into a single entity named "Other." Table 1 summarizes the number of recorded time series at each level, highlighting the dataset's hierarchical structure. To speed up the training process and reduce the computing complexity of

Table 1 Number of time series per level of all hierarchies for the DJIA market value dataset

Level	Series Count
Level 0 (Top Level)	1
Level 1 (Middle level)	6
Level 2 (Bottom level)	30
Total	37

big numerical values, all market value data points were scaled down to trillions of dollars. This rescaling technique accelerated forecasting model training while maintaining numerical stability and efficiency throughout the study. Figure 3 displays the dataset's hierarchical structure.

Table 2 shows the DJIA component companies, their corresponding industries, Ticker symbol and the outstanding shares and Table 3 shows the summary statistics of the DJIA companies (Fig. 1).

Explanation:

- **A:** Conglomerate industry, which includes stocks A1 (MMM), and A2 (HON).
- **B:** Financial Services industry, which includes stocks B1 (GS), B2 (JPM), B3 (AXP), and B4 (V).
- **C:** Biopharmaceutical industry, which includes stocks C1 (AMGN), C2 (MRK), and C3 (JNJ).
- **D:** Retailing industry, which includes stocks D1 (WMT) and D2 (AMZN).
- **E:** Information Technology industry, which includes stocks E1 (AAPL), E2 (MSFT), E3 (CRM), E4 (IBM), and E5 (CSCO).
- **F:** Other industries, which include stocks F1 (BA), F2 (CAT), F3 (CVX), F4 (KO), F5 (DIS), F6 (DOW), F7 (HD), F8 (MCD), F9 (NKE), F10 (PG), F11 (UNH), F12 (VZ), F13 (TVR), and F14 (INTC).

In Fig. 2, we show the time series plot for the aggregate market value of the entire DJIA Market Index and the respective industries from January 4, 2021, to May 31, 2024.

In Fig. 3, we show the time series plot for the market value of each company in the DJIA Market Index, grouped by industry.

3.3 Hierarchical Time Series (HTS)

The hierarchical structure of the market value for the DJIA market index, which encompasses 30 stocks across six industries, can be divided into three levels: the top level representing the total aggregate market value, the middle level representing the market values of the six industries, and the bottom level representing the market values of the 30 individual stocks. At the top level, we have the total aggregate market value, denoted as f_t . This value represents the combined market value of all industries and stocks at time t . The middle level breaks down the total market value into six industries: Conglomerate (A), Financial Services (B), Biopharmaceutical (C), Retailing (D), Information Technology (E), and Other industries (F). Therefore, each industry's market value at time t is represented as $f_{A,t}$, $f_{B,t}$, $f_{C,t}$, $f_{D,t}$, $f_{E,t}$, and $f_{F,t}$.

The bottom level further breaks down each industry's market value into individual stocks or companies. The market values of these stocks at time t are denoted by $f_{I,j,t}$, where I represents the industry and j represents the stock within that industry. For example, industry A includes stocks $f_{A1,t}$ and $f_{A2,t}$; industry B includes stocks $f_{B1,t}$, $f_{B2,t}$, $f_{B3,t}$, and $f_{B4,t}$; and so on, with industry F having 14 stocks from $f_{F1,t}$ to $f_{F14,t}$. To maintain this hierarchical structure, we define how the values at different levels are summed up. The total aggregate market value f_t is the sum of the

Table 2 DJIA component companies their respective Industry, Ticker Symbol and the number of outstanding shares as of May 31, 2024

No.	Company	Industry	Ticker Symbol	Outstanding Shares
1	3M	Conglomerate	MMM	553,361,024
2	American Express	Financial Services	AXP	719,302,976
3	Amgen	Biopharmaceutical	AMGN	536,435,008
4	Amazon	Retailing	AMZN	10,406,599,680
5	Apple	Information Technology	AAPL	15,334,099,968
6	Boeing	Aerospace and Defense	BA	613,884,032
7	Caterpillar	Construction and Mining	CAT	489,052,992
8	Chevron	Petroleum Industry	CVX	1,828,800,000
9	Cisco	Information Technology	CSCO	4,028,809,984
10	Coca-Cola	Drink Industry	KO	4,307,959,808
11	Disney	Broadcasting and Entertainment	DIS	1,823,040,000
12	Dow	Chemical Industry	DOW	703,206,016
13	Goldman Sachs	Financial Services	GS	322,463,008
14	Home Depot	Home Improvement	HD	991,614,016
15	Honeywell	Conglomerate	HON	651,185,984
16	IBM	Information Technology	IBM	918,603,008
17	Intel	Semiconductor Industry	INTC	4,256,869,888
18	Johnson and Johnson	Pharmaceutical Industry	JNJ	2,406,680,064
19	JPMorgan Chase	Financial Services	JPM	2,871,670,016
20	McDonald's	Food Industry	MCD	720,681,984
21	Merck	Pharmaceutical Industry	MRK	2,532,809,984
22	Microsoft	Information Technology	MSFT	7,432,309,760
23	Nike	Clothing Industry	NKE	1,211,459,968
24	Procter and Gamble	Fast-Moving Consumer Goods	PG	2,360,140,032
25	Salesforce	Information Technology	CRM	969,000,000
26	Travelers	Insurance	TRV	228,992,992
27	United-Health Group	Managed Health Care	UNH	920,385,024
28	Verizon	Telecommunications Industry	VZ	4,209,250,048
29	Visa	Financial Services	V	1,574,150,016
30	Walmart	Retailing	WMT	8,043,539,968

Table 3 Statistical summary of stock market value for all 30 companies indexed in DJIA Market. The values are scaled into Trillion

Stock	mean	std	min	25%	50%	75%	max	kurtosis	skewness
AAPL	2.4481	0.3220	1.7843	2.2006	2.4426	2.6931	3.3224	-0.9247	0.0462
AMGN	0.1337	0.0137	0.1067	0.1238	0.1316	0.1425	0.1741	0.0038	0.6385
AMZN	1.4898	0.2986	0.8515	1.2559	1.5442	1.7429	1.9721	-1.0664	-0.3685
AXP	0.1210	0.0182	0.0821	0.1098	0.1178	0.1267	0.1748	1.1203	1.0670
BA	0.1227	0.0193	0.0711	0.1112	0.1263	0.1343	0.1653	-0.0803	-0.5125
CAT	0.1159	0.0230	0.0794	0.1004	0.1092	0.1237	0.1855	0.7565	1.1749
CRM	0.2121	0.0436	0.1243	0.1813	0.2087	0.2429	0.3071	-0.7188	0.1022
CSCO	0.2038	0.0186	0.1582	0.1910	0.2022	0.2171	0.2577	-0.3074	0.1671
CVX	0.2614	0.0497	0.1549	0.2088	0.2816	0.2977	0.3439	-1.0676	-0.5652
DIS	0.2274	0.0664	0.1446	0.1721	0.1990	0.2944	0.3681	-1.2145	0.6025
DOW	0.0399	0.0040	0.0305	0.0370	0.0397	0.0424	0.0499	-0.3359	0.2758
GS	0.1141	0.0127	0.0855	0.1048	0.1117	0.1235	0.1517	-0.0852	0.5049
HD	0.3175	0.0326	0.2488	0.2935	0.3146	0.3322	0.4127	0.2923	0.7087
HON	0.1316	0.0095	0.1087	0.1251	0.1307	0.1382	0.1525	-0.5083	0.1874
IBM	0.1285	0.0167	0.1041	0.1180	0.1245	0.1320	0.1817	1.3898	1.3662
INTC	0.1800	0.0471	0.1060	0.1357	0.1813	0.2206	0.2906	-1.1434	0.2343
JNJ	0.3959	0.0207	0.3475	0.3827	0.3948	0.4116	0.4477	-0.4996	-0.0530
JPM	0.4269	0.0624	0.2928	0.3851	0.4243	0.4608	0.5881	-0.0799	0.4110
KO	0.2535	0.0170	0.2074	0.2413	0.2577	0.2667	0.2852	-0.3513	-0.5972
MCD	0.1866	0.0166	0.1476	0.1738	0.1867	0.1990	0.2166	-0.7306	-0.1555
MMM	0.0641	0.0170	0.0396	0.0490	0.0586	0.0817	0.0959	-1.2956	0.4044
MRK	0.2430	0.0458	0.1744	0.1975	0.2352	0.2775	0.3342	-1.1777	0.2765
MSFT	2.2526	0.4249	1.5775	1.8922	2.1767	2.4874	3.3324	-0.4636	0.6752
NKE	0.1487	0.0282	0.1007	0.1273	0.1437	0.1656	0.2150	-0.6555	0.5762
PG	0.3468	0.0237	0.2883	0.3303	0.3459	0.3644	0.3978	-0.6007	-0.1077
TRV	0.0396	0.0047	0.0308	0.0361	0.0387	0.0419	0.0529	0.3695	0.9636
UNH	0.4370	0.0505	0.2985	0.4137	0.4505	0.4742	0.5110	0.1228	-0.9514
V	0.3597	0.0379	0.2796	0.3333	0.3555	0.3769	0.4571	-0.1070	0.6240
VZ	0.1906	0.0355	0.1291	0.1595	0.1766	0.2255	0.2505	-1.5249	0.1183
WMT	0.3994	0.0421	0.3172	0.3741	0.3898	0.4199	0.5437	1.2515	1.0382

market values of all six industries, and each industry's market value is the sum of the market values of its constituent stocks.

To efficiently represent and compute these summations, we use vector and matrix notation. Let f_t be the vector of the total and industry market values and individual stock market value, and f_b be the vector of only the individual stock market values. The relationship between the total aggregate market value vector f_t and the market values of the individual stocks f_b is given by the summing matrix S . This matrix S captures the hierarchical structure and ensures that the aggregate market values are accurately computed from the base stock values. The matrix equation $f_t = S f_b$ concisely represents this hierarchical aggregation. The summation matrix S is constructed to reflect the relationships between the total market value, industry market values, and individual stock market values, allowing for efficient computation and clear representation of the market value hierarchy.

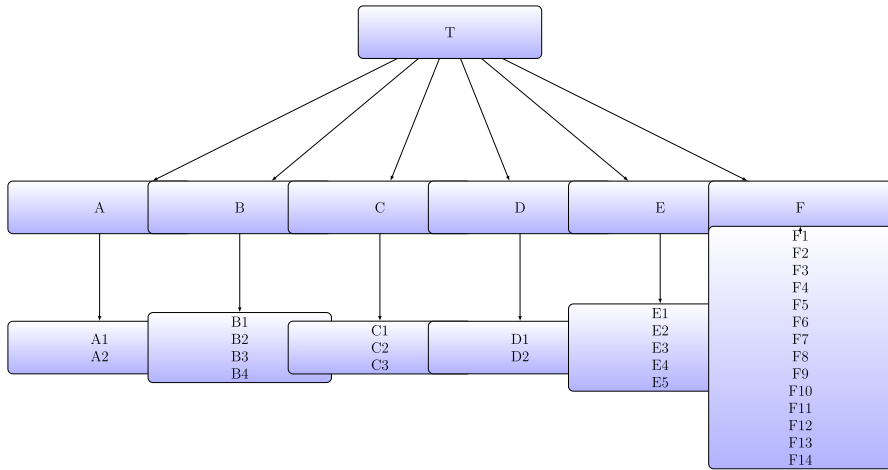


Fig. 1 Hierarchical structure of the DJIA market value dataset

The total aggregate market value f_t is the sum of the market values of all six industries:

$$f_t = f_{A,t} + f_{B,t} + f_{C,t} + f_{D,t} + f_{E,t} + f_{F,t} \quad (1)$$

Each industry's market value is the sum of the market values of its constituent stocks. For example:

$$f_{A,t} = f_{A1,t} + f_{A2,t} \quad (2)$$

$$f_{B,t} = f_{B1,t} + f_{B2,t} + f_{B3,t} + f_{B4,t} \quad (3)$$

$$f_{C,t} = f_{C1,t} + f_{C2,t} + f_{C3,t} \quad (4)$$

$$f_{D,t} = f_{D1,t} + f_{D2,t} \quad (5)$$

$$f_{E,t} = f_{E1,t} + f_{E2,t} + f_{E3,t} + f_{E4,t} + f_{E5,t} \quad (6)$$

$$f_{F,t} = f_{F1,t} + f_{F2,t} + \cdots + f_{F14,t} \quad (7)$$

To efficiently represent and compute these summations, we use vector and matrix notation.

The relationship between the total aggregate market value vector f_t and the market values of the individual stocks f_b through the summing matrix S is given by:

$$f_t = S f_b \quad (8)$$

where

$$\begin{bmatrix} f_t \\ f_{A,t} \\ f_{B,t} \\ f_{C,t} \\ f_{D,t} \\ f_{E,t} \\ f_{F,t} \\ f_{A1,t} \\ f_{A2,t} \\ f_{B1,t} \\ f_{B2,t} \\ f_{B3,t} \\ f_{B4,t} \\ f_{C1,t} \\ f_{C2,t} \\ f_{C3,t} \\ f_{D1,t} \\ f_{D2,t} \\ f_{E1,t} \\ f_{E2,t} \\ f_{E3,t} \\ f_{E4,t} \\ f_{E5,t} \\ f_{F1,t} \\ f_{F2,t} \\ \vdots \\ f_{F13,t} \\ f_{F14,t} \end{bmatrix} = S \times \begin{bmatrix} f_{A1,t} \\ f_{A2,t} \\ f_{B1,t} \\ f_{B2,t} \\ f_{B3,t} \\ f_{B4,t} \\ f_{C1,t} \\ f_{C2,t} \\ f_{C3,t} \\ f_{D1,t} \\ f_{D2,t} \\ f_{E1,t} \\ f_{E2,t} \\ f_{E3,t} \\ f_{E4,t} \\ f_{E5,t} \\ f_{F1,t} \\ f_{F2,t} \\ \vdots \\ f_{F13,t} \\ f_{F14,t} \end{bmatrix}$$

The summing matrix S is used to aggregate the stock values into industry values and then the industry values into the total market value. For our hierarchical structure, S is defined as:

The summing matrix S :

$$S = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & \dots & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & \dots & 1 & 1 \end{bmatrix}$$

I_m

Where f_t is an n -dimensional vector of all observations at time t , S be the $n \times m$ sum matrix, and f_b is an m -dimensional vector of all observations at the lower level of the series at time t . Where I_m denotes the $m \times m$ identity matrix with $m = 30$. The values of one in the matrix S indicate the existence of stocks, while zeros do not, to generate the aggregated market value for the DJIA index and by industry. Thus, the first row of this matrix represents the total market value of the DJIA composed of thirty stocks; the second row represents the Conglomerate industry composed of two corresponding stocks; the third row represents the Financial Services industry composed of four corresponding stocks; the fourth row represents the Biopharma-

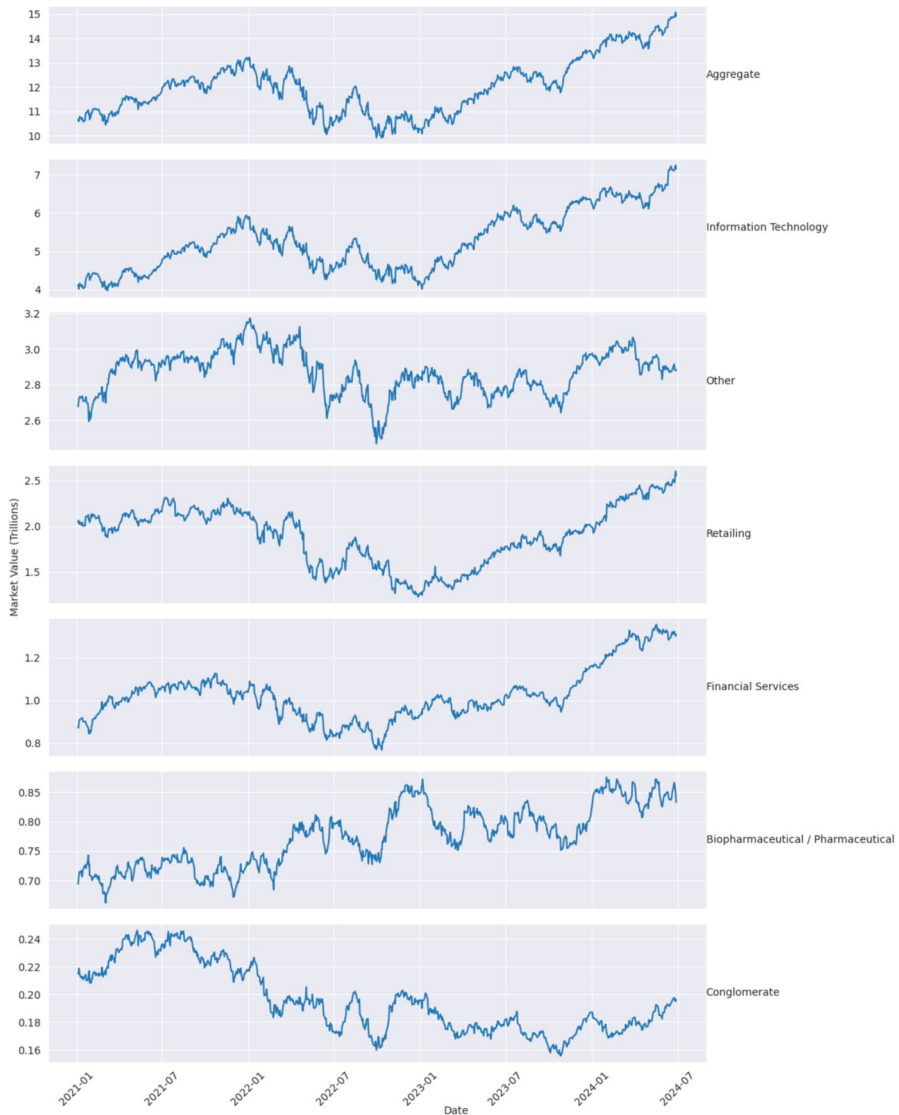


Fig. 2 Time series plot for aggregate market value in Trillion US dollars of the entire DJIA Market Index and the Industries from January 4, 2021, to May 31, 2024

ceutical industry composed of three corresponding stocks; the fifth row represents the Retailing industry composed of two corresponding stocks; the sixth row represents the Information Technology industry composed of five corresponding stocks; and, finally, the seventh row represent the Other industries composed of fourteen corresponding stocks. The rows of the identity matrix I_m characterize each corresponding stock to allow the construction of aggregated market values.

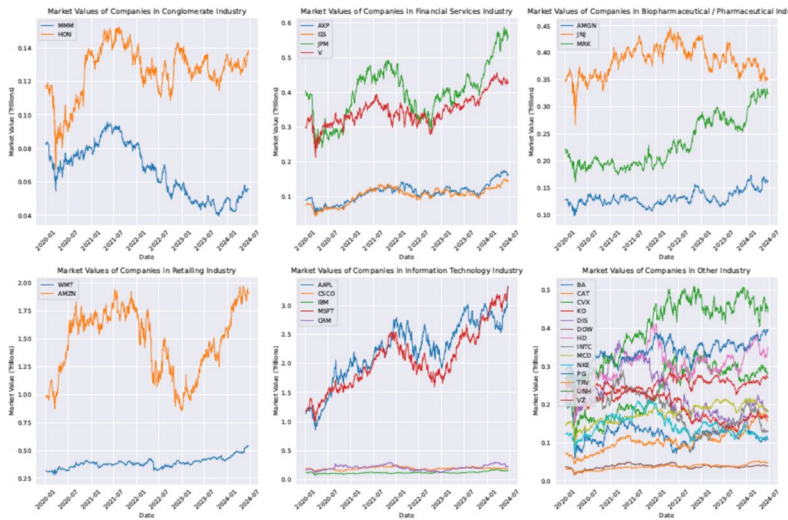


Fig. 3 Time series plot for market value in Trillion US dollars of each company (companies grouped per industry) from for all trading days

3.4 Forecasting Methods for Hierarchical Time Series

All forecasting approaches can be represented using a single matrix notation:

$$\hat{f}_t = SG\hat{f}_b \quad (9)$$

where $\hat{f}_t$ represents the coherent forecasts, S is the sum matrix, $\hat{f}_b$ is the base forecasts for all time series in the hierarchy, and G is the matrix mapping the base forecasts, which differ in each approach. We describe in brief the various ways to create coherent forecasts for a hierarchical time series that are explored in this work.

3.4.1 Bottom-Up (BU) Approach

The Bottom-Up (BU) technique entails creating forecasts for each time series at the lowest granular level, which are then aggregated to produce forecasts for higher levels. This strategy takes advantage of all available comprehensive information because projections are created at the lowest level. However, data at the lowest level might be very noisy, resulting in erroneous forecasts. This method can be stated in the generic form of Eq. 9, as follows:

$$G = [0_{m \times (n-m)} \quad I_m] \quad (10)$$

where $0_{m \times (n-m)}$ is a null matrix and I_m is the identity matrix. The matrix G , which maps the base forecasts, is an identity matrix for the section corresponding to the

lower-level forecasts. These individual forecasts are directly employed to produce coherent forecasts.

The Bottom-Up technique is useful when the lower levels of the hierarchy make a significant contribution to the upper levels or when the data quality at the lower levels is better. This approach has the advantage of forecasting at the bottom level of the structure, preventing information loss due to aggregation (Hyndman & Khandakar, 2008). However, bottom-level data can be noisy and difficult to predict and forecast.

3.4.2 Top-Down Forecasting Methods

The top-down (TD) approach is utilized in forecasting with strictly hierarchical aggregation structures, rather than grouped structures. This method begins by making forecasts for the total market value at the top level, $\hat{f}_t$, and then disaggregates these forecasts down the hierarchy to obtain forecasts for all series at the bottom level.

Given the total forecast $\hat{f}_t$, the bottom-level forecasts for each stock $\hat{f}_{i,t}$ are obtained by multiplying the total forecast by the respective disaggregation proportions p_i .

For the Conglomerate industry, which includes stocks A1 and A2:

$$\hat{f}_{A1,t} = p_{A1}\hat{f}_t \quad (11)$$

$$\hat{f}_{A2,t} = p_{A2}\hat{f}_t \quad (12)$$

For the Financial Services industry, which includes stocks B1, B2, B3, and B4:

$$\hat{f}_{B1,t} = p_{B1}\hat{f}_t \quad (13)$$

$$\hat{f}_{B2,t} = p_{B2}\hat{f}_t \quad (14)$$

$$\hat{f}_{B3,t} = p_{B3}\hat{f}_t \quad (15)$$

$$\hat{f}_{B4,t} = p_{B4}\hat{f}_t \quad (16)$$

For the Biopharmaceutical industry, which includes stocks C1, C2, and C3:

$$\hat{f}_{C1,t} = p_{C1}\hat{f}_t \quad (17)$$

$$\hat{f}_{C2,t} = p_{C2}\hat{f}_t \quad (18)$$

$$\hat{f}_{C3,t} = p_{C3}\hat{f}_t \quad (19)$$

For the Retailing industry, which includes stocks D1 and D2:

$$\hat{f}_{D1,t} = p_{D1} \hat{f}_t \quad (20)$$

$$\hat{f}_{D2,t} = p_{D2} \hat{f}_t \quad (21)$$

For the Information Technology industry, which includes stocks E1, E2, E3, E4, and E5:

$$\hat{f}_{E1,t} = p_{E1} \hat{f}_t \quad (22)$$

$$\hat{f}_{E2,t} = p_{E2} \hat{f}_t \quad (23)$$

$$\hat{f}_{E3,t} = p_{E3} \hat{f}_t \quad (24)$$

$$\hat{f}_{E4,t} = p_{E4} \hat{f}_t \quad (25)$$

$$\hat{f}_{E5,t} = p_{E5} \hat{f}_t \quad (26)$$

For other industries, which include stocks F1 through F14:

$$\hat{f}_{F1,t} = p_{F1} \hat{f}_t \quad (27)$$

$$\hat{f}_{F2,t} = p_{F2} \hat{f}_t \quad (28)$$

$$\hat{f}_{F3,t} = p_{F3} \hat{f}_t \quad (29)$$

$$\hat{f}_{F4,t} = p_{F4} \hat{f}_t \quad (30)$$

$$\hat{f}_{F5,t} = p_{F5} \hat{f}_t \quad (31)$$

$$\hat{f}_{F6,t} = p_{F6} \hat{f}_t \quad (32)$$

$$\hat{f}_{F7,t} = p_{F7} \hat{f}_t \quad (33)$$

$$\hat{f}_{F8,t} = p_{F8} \hat{f}_t \quad (34)$$

$$\hat{f}_{F9,t} = p_{F9} \hat{f}_t \quad (35)$$

$$\hat{f}_{F10,t} = p_{F10} \hat{f}_t \quad (36)$$

$$\hat{f}_{F11,t} = p_{F11} \hat{f}_t \quad (37)$$

$$\hat{f}_{F12,t} = p_{F12} \hat{f}_t \quad (38)$$

$$\hat{f}_{F13,t} = p_{F13} \hat{f}_t \quad (39)$$

$$\hat{f}_{F14,t} = p_{F14} \hat{f}_t \quad (40)$$

The general formula for the bottom-level forecasts in terms of disaggregation proportions is:

$$\hat{f}_{i,t} = p_i \hat{f}_t \quad (41)$$

Using matrix notation, let p be the vector of proportions and $\hat{f}_t$ be the vector of bottom-level forecasts. We can write:

$$\hat{f}_t = p \hat{f}_t \quad (42)$$

Where p is a 30×1 vector containing the proportions p_i for each of the 30 stocks, and $\hat{f}_t$ is a 30×1 vector of the bottom-level forecasts.

To effectively represent and compute these summations, we use vector and matrix notation. Here, f_t represents the total aggregate market value, and f_b represents the market values of the individual stocks. The relationship between f_t and f_b is given by:

$$f_t = S f_b \quad (43)$$

where S is the summing matrix. Let p be the vector of disaggregation proportions $(p_1, p_2, \dots, p_{30})$, which will be used to distribute the forecast of the total market value to obtain the forecasts for all individual stocks at the bottom level. The h -step-ahead predictions for the bottom-level series can be written as:

$$f_{b,t+h} = p \cdot f_{t+h} \quad (44)$$

In matrix form, this can be represented as:

$$f_b = S p \cdot f_t \quad (45)$$

The main top-down models stipulate disaggregation proportions according to the historical proportions of the data. Some common top-down methods include the Top-Down Gross-Sohl Method A (TDGSA), Top-Down Gross-Sohl Method F (TDGSF), and Top-Down Forecast Proportions (TDFP). These methods allow for accurate and consistent forecasting across different levels of the hierarchy (Gross & Sohl, 1990).

1. **Proportions of Historical Averages** The proportions of historical averages, designated by p_j , are determined as:

$$p_j = \frac{\sum_{t=1}^T f_{j,t}/T}{\sum_{t=1}^T f_t/T}, \quad \text{for } j = 1, \dots, m \quad (46)$$

The proportion p_j represents the average historical value of the bottom-level series $f_{j,t}$ compared to the average value of the whole aggregate f_t .

2. **Average Historical Proportions** The average historical percentage, indicated by p_j , is determined as follows:

$$p_j = \frac{1}{T} \sum_{t=1}^T \frac{f_{j,t}}{f_t}, \quad \text{for } j = 1, \dots, m \quad (47)$$

Each percentage p_j is the average of the historical proportions of the bottom-level series $f_{j,t}$ throughout the period $t = 1, \dots, T$ relative to the entire aggregate f_t .

3. **Forecast Proportions** Forecast proportions are generated using forecasts rather than historical data. For a K -level hierarchy, the general procedure for obtaining the forecast proportions is given by:

$$p_j = \prod_{\ell=0}^{K-1} \frac{\hat{f}_{j,h}^{(\ell)} \hat{S}_{j,h}^{(\ell+1)}}{\hat{S}_{j,h}^{(\ell)}}, \quad \text{for } j = 1, \dots, m \quad (48)$$

where $\hat{f}_{j,h}^{(\ell)}$ is the h -step-ahead initial forecast of the series corresponding to the node which is ℓ levels above j , and $\hat{S}_{j,h}^{(\ell)}$ is the sum of the h -step-ahead initial forecasts below the node that is ℓ levels above node j and are directly connected to that node. These forecast proportions disaggregate the h -step-ahead initial forecast of the total series to get h -step-ahead coherent forecasts of the bottom-level series.

3.4.3 Middle-out Approach

The middle-out strategy combines both bottom-up and top-down approaches. The aim is to select a 'middle' level (the level between the top and bottom of the hierarchy) and anticipate all series at that level. The bottom-up technique is then utilized to construct coherent forecasts for series above the specified intermediate level, which are aggregated upwards. The top-down technique generates coherent forecasts by disaggregating middle-level projections downwards for the series below the middle level.

3.4.4 Optimal Reconciliation Method

The MinT (Minimum Trace) optimal reconciliation method, introduced by Wickramasuriya et al. (2018), seeks to minimize the total forecast variance to achieve coherent forecasts.

Generally, coherent forecasting methods for hierarchical or grouped structures can be expressed as:

$$\tilde{f}_h = SG\hat{f}_h, \quad (49)$$

where G is a matrix that maps the base forecasts $\hat{f}_h$ to the bottom level, and the summing matrix S aggregates these forecasts to produce a set of coherent forecasts $\tilde{f}_h$.

To ensure unbiased forecasts, if the base forecasts $\hat{f}_h$ are unbiased, then the coherent forecasts $\tilde{f}_h$ will also be unbiased if $SGS = S$. However, no top-down method satisfies this constraint, potentially leading to biased coherent forecasts.

Wickramasuriya et al. (2018) derived the variance-covariance matrix of the h -step-ahead coherent forecast errors:

$$V_h = SGW_hG'S', \quad (50)$$

where $W_h = \text{Var}[f_{T+h} - \tilde{f}_h]$ is the variance-covariance matrix of the base forecast errors.

The goal is to find a matrix G that minimizes the error variances of the coherent forecasts. The matrix G that minimizes the trace of V_h subject to $SGS = S$ is given by:

$$G = (S'W_h^{-1}S)^{-1}S'W_h^{-1}. \quad (51)$$

The optimally reconciled forecasts are then obtained as:

$$\tilde{f}_h = S(S'W_h^{-1}S)^{-1}S'W_h^{-1}\hat{f}_h. \quad (52)$$

The MinT optimal reconciliation approach leverages all available information within a hierarchical or grouped structure to generate forecasts, revealing features of the data that may be obscured at other levels.

3.5 Recurrent Neural Networks (RNNs)

RNN excel at processing sequential data due to their recurrent connections, which enable the retention of information from previous time steps. This capability allows RNNs to effectively handle tasks where context and order matter, such as predicting words in a sentence or forecasting future values in time series data like stock prices. By feeding the output from the previous step as input to the current step, RNNs maintain an internal state that captures past information, aiding in understanding and predicting sequences. This memory mechanism allows RNNs to capture patterns and relationships over time, making them particularly suitable for tasks where historical data significantly influences future outcomes.

The mathematical foundation of RNNs is encapsulated in a few key equations. The hidden state at time t , denoted as h_t , is computed using the current input x_t , the previous hidden state h_{t-1} , and associated weight matrices and biases. The operation is defined as:

$$h_t = \phi(W_{hh}h_{t-1} + W_{xh}x_t + b_h) \quad (53)$$

where ϕ is the activation function, typically a non-linear function such as tanh or ReLU. The output vector y_t at time step t is then given by:

$$y_t = W_{hy}h_t + b_y \quad (54)$$

Here, W_{hh} , W_{xh} , and W_{hy} are weight matrices, while b_h and b_y are bias vectors. Despite their strengths, traditional RNNs face challenges with long-term dependencies due to the vanishing gradient problem, where the influence of earlier information diminishes over time. This issue has led to the development of advanced RNN variants like LSTM networks and GRU networks, which are designed to more effectively capture and retain long-range dependencies by incorporating gating mechanisms (Lipton et al., 2015; Bullinaria, 2013; Yu et al., 2019). Figure 4 show a RNN layout

3.6 Long Short-Term Memory (LSTM)

LSTM is a specific type of RNN architecture created to address the vanishing gradient problem found in standard RNNs, which hampers their ability to learn and remember information across lengthy sequences. LSTMs are particularly useful for modeling and forecasting sequential data, making them ideal for applications like time series forecasting, natural language processing, and speech recognition (Wang et al., 2019; Karevan & Suykens, 2020; Hochreiter & Schmidhuber, 1997a; Wan et al., 2020; Bala et al., 2019).

An LSTM unit is composed of several key components: a cell state C_t , which acts as the memory of the network, and three gates - the input gate i_t , the forget gate f_t , and the output gate o_t . These gates control the flow of information into and out of the cell state, enabling the LSTM to selectively update, retain, or discard information as time progresses.

The behavior of an LSTM unit is described by the following equations:

The input gate i_t determines how much of the new information to let into the cell state:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (55)$$

The forget gate f_t decides how much of the previous cell state C_{t-1} to retain:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (56)$$

The output gate o_t regulates the information to be outputted from the cell state:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (57)$$

The candidate cell state $\tilde{C}_t$ is calculated based on the current input and the previous hidden state:

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (58)$$

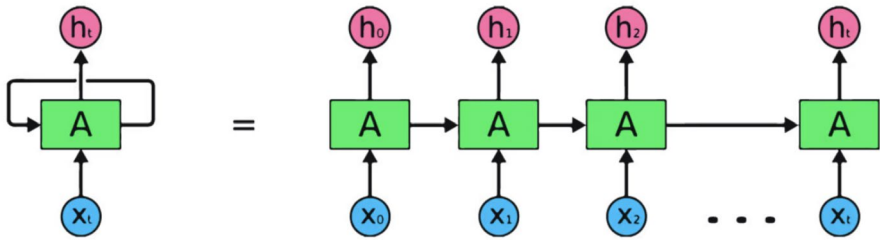


Fig. 4 RNN Layout

The cell state C_t is updated by combining the previous cell state C_{t-1} , modified by the forget gate, and the candidate cell state $\tilde{C}_t$, adjusted by the input gate:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (59)$$

The hidden state h_t is then determined by applying the output gate to the cell state:

$$h_t = o_t \cdot \tanh(C_t) \quad (60)$$

Here, i_t , f_t , and o_t represent the input, forget, and output gates, respectively, while $\tilde{C}_t$ is the candidate cell state, C_t is the cell state, and h_t is the hidden state. Additionally, x_t is the input at time step t , and W_i , W_f , W_o , W_c , b_i , b_f , b_o , and b_c are the weight matrices and bias vectors associated with the gates. The sigmoid activation function is denoted by σ , and the concatenation of h_{t-1} and x_t is represented as $[h_{t-1}, x_t]$. Figure 5 show a LSTM layout.

3.7 Gated Recurrent Unit (GRU)

GRU is a RNN architecture developed to overcome some of the challenges associated with traditional RNNs, such as problems in learning long-range dependencies and the vanishing gradient problem. GRU simplifies the architecture of the LSTM unit while maintaining its capability to capture long-term dependencies in sequential data (Shen et al., 2018; Dey & Salem, 2017; Peng et al., 2020; Wu et al., 2019). The GRU unit is governed by the following equations:

The update gate z_t controls the extent to which the past state information is retained and how much new information is incorporated:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (61)$$

The reset gate r_t determines how much of the previous state to forget:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (62)$$

The candidate activation $\tilde{h}_t$ represents the new candidate hidden state based on the current input and the reset gate:

$$\tilde{h}_t = \tanh(W \cdot [r_t \cdot h_{t-1}, x_t] + b) \quad (63)$$

The hidden state h_t is updated by blending the previous hidden state h_{t-1} with the candidate activation $\tilde{h}_t$, weighted by the update gate:

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \quad (64)$$

Figure 6 show a GRU layout

3.8 Experimental Setup

Regarding the dataset, two distinct forecast horizons were considered: short-term and long-term. For short-term forecasting, the 115 last days of data were used as the test set, representing a forecast horizon of approximately half a trading year. This shorter horizon allows for more immediate, short-term predictions, where the model's ability to capture recent patterns and trends is crucial. The remaining data (all but the last 115 days) was used for training, providing the model with historical data to learn the underlying patterns and make accurate predictions.

For long-term forecasting, the 230 last days of data were used as the test set, representing a longer-term forecast horizon. This horizon spans nearly a full trading year, enabling the model to capture longer-term trends and fluctuations that may be influenced by different factors compared to short-term movements. In this case, the remaining data was used for training, allowing the model to leverage a larger portion of historical data to make predictions over a more extended period.

This division of data into short-term and long-term forecast horizons ensured that the models were evaluated on both short-term (115 days) and long-term (230 days) forecasting tasks, which are critical for understanding their performance across different time scales.

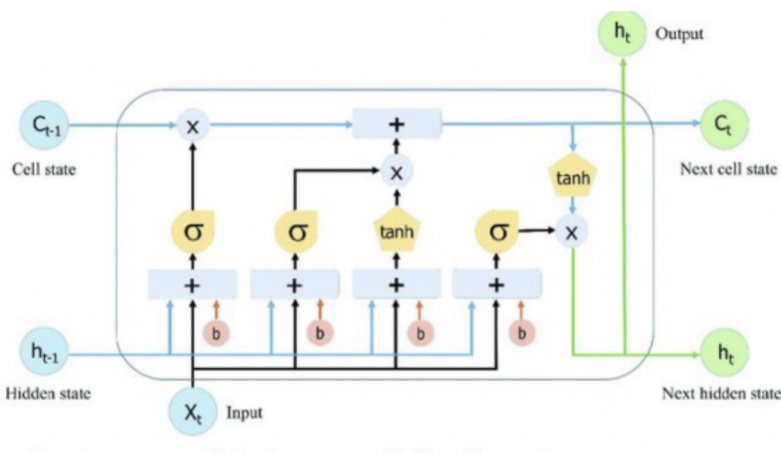


Fig. 5 LSTM Layout

Grid search was conducted using the Mean Squared Error loss function to determine the optimal hyperparameters for the models. We first considered the Top series which represent the aggregate market value in processing of getting training parameters for our models using LSTM model.

The grid search considered different combinations of training parameters. The number of epochs tested were 50 and 100, batch sizes were 32 and 64, learning rates were 0.001 and 0.0001, and the optimizers used were Adam and RMSprop. Model architectures included two layers with configurations of 128 and 64 neurons, as well as 100 and 50 neurons. For all models, dropout regularization was applied at a rate of 0.2 to prevent overfitting. The optimal parameters identified were as follows: a batch size of 32, 100 epochs, a learning rate of 0.001, and the Adam optimizer. Dropout was applied at a rate of 0.2 to prevent overfitting, and early stopping was implemented to halt training if the validation loss did not improve for a certain number of epochs. These parameters were set consistently across all 37 series at different hierarchical levels.

To ensure consistency and fair comparison, the same parameters and procedures were applied across all hierarchical levels for training and testing the Simple RNN, LSTM, and GRU models. This approach allowed for a comprehensive evaluation of the forecasting performance of each model, enabling meaningful comparisons across the different levels of the hierarchical time series data.

The dataset was initially scaled using the MinMaxScaler. Subsequently, the Simple RNN, LSTM, and GRU models were built using the Sequential API from TensorFlow. After training, predictions were made for both the training and testing datasets,

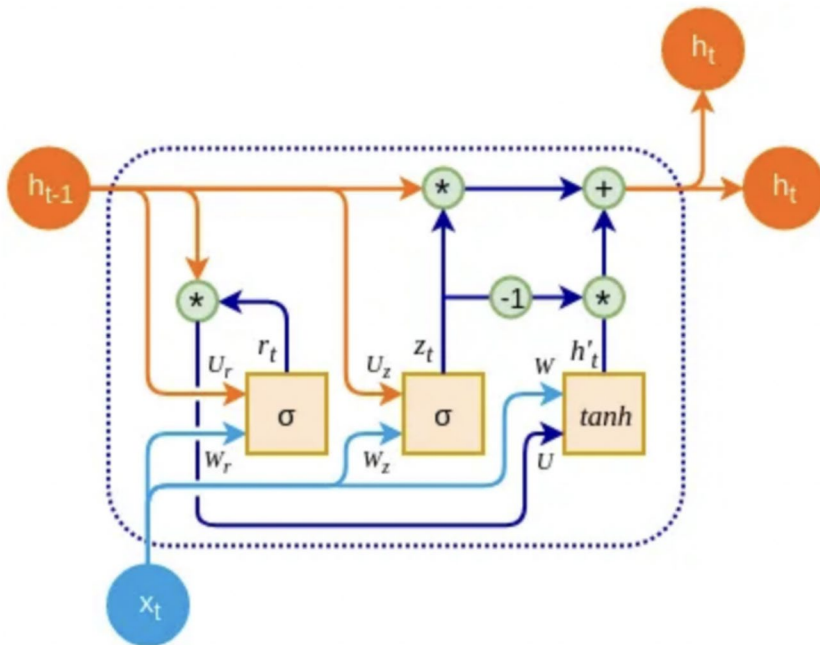


Fig. 6 GRU layout

and an inverse transformation was applied to obtain the actual predicted values. The predictions were then stored in a data frame for further analysis.

We used several selected reconciliation methods to compare hierarchical time series data. These methods include BottomUp (BU), TopDown using forecast proportions (TD-Fp), MiddleOut with forecast proportions (MO-Fp), MinTrace with ordinary least squares (MT-Ols), MinTrace with weighted least squares (MT-Wls), and Optimal Combination using ordinary least squares (OC-Ols). Reconciliation Was done using the Nixtla package in Python (Olivares et al., 2022). Figure 7 show the flow chart of the experimental design.

3.9 Model Evaluation Metrics

We discuss metrics used to examine forecasting accuracy in this section.

3.9.1 Average RMSE (AveRMSE)

First, to calculate the RMSE for each individual time series k , we use the formula:

$$\text{RMSE}_k = \sqrt{\frac{1}{n_k} \sum_{i=1}^{n_k} (y_{i,k} - \hat{y}_{i,k})^2} \quad (65)$$

where n_k is the number of observations in the k -th time series, $y_{i,k}$ is the actual value of the i -th observation in the k -th time series, and $\hat{y}_{i,k}$ is the predicted value of the i -th observation in the k -th time series. To find the average RMSE at level j , we average the RMSE values across all time series at that level using the formula:

$$\text{Average RMSE}_j = \frac{1}{n_j} \sum_{k=1}^{n_j} \sqrt{\frac{1}{n_k} \sum_{i=1}^{n_k} (y_{i,k} - \hat{y}_{i,k})^2} \quad (66)$$

where n_j is the number of time series at level j .

3.9.2 Average MAPE (AveMAPE)

Similarly, to calculate the MAPE for each individual time series k , we use the formula:

$$\text{MAPE}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \left| \frac{y_{i,k} - \hat{y}_{i,k}}{y_{i,k}} \right| \times 100\% \quad (67)$$

where n_k is the number of observations in the k -th time series, $y_{i,k}$ is the actual value of the i -th observation in the k -th time series, and $\hat{y}_{i,k}$ is the predicted value of the i -th observation in the k -th time series. To find the average MAPE at level j , we average the MAPE values across all time series at that level using the formula:

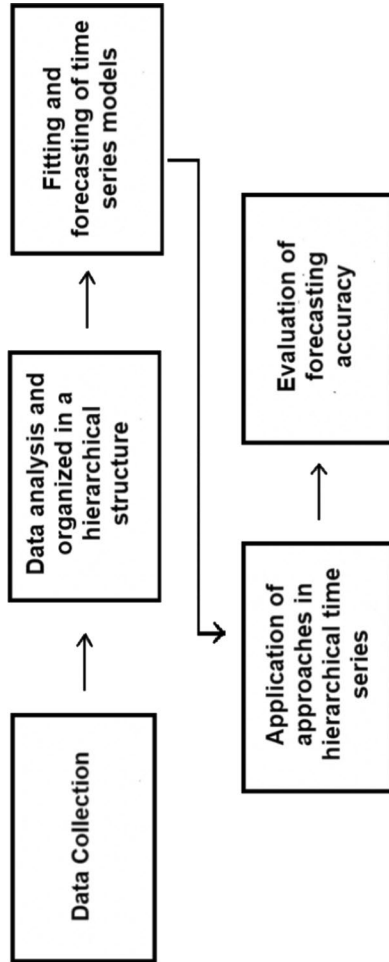


Fig. 7 Flowchart of the steps of methodology

$$\text{Average MAPE}_j = \frac{1}{n_j} \sum_{k=1}^{n_j} \left(\frac{1}{n_k} \sum_{i=1}^{n_k} \left| \frac{y_{i,k} - \hat{y}_{i,k}}{y_{i,k}} \right| \times 100\% \right) \quad (68)$$

where n_j is the number of time series at level j .

3.9.3 Average Relative Root Mean Squared Error (AvgRelRMSE)

The Average Relative Root Mean Squared Error (AvgRelRMSE), as suggested by Davydenko and Fildes (2013), measures the effectiveness of reconciliation methods by comparing them to base forecasts. It is computed as the geometric mean of the ratios of Root Mean Squared Errors (RMSEs) before and after reconciliation across all time series. Mathematically, AvgRelRMSE is expressed as shown below:

$$\text{AvgRelRMSE} = \left(\prod_{i=1}^m r_i \right)^{\frac{1}{m}} ; \quad r_i = \frac{\text{RMSERec}_i}{\text{RMSEBase}_i} \quad (69)$$

In this formula, RMSEBase_i denotes the RMSE of the base forecasts for time series i , while RMSERec_i represents the RMSE for the same time series i after applying reconciliation. The variable m indicates the total number of time series. A key benefit of AvgRelRMSE is its ability to quantify the percentage improvement of reconciliation methods over base forecasts, calculated as $(1 - \text{AvgRelRMSE}) \times 100\%$. Similarly, AvgRelMAPE was calculated using an analogous approach.

4 Experimental Results and Discussion

Table 4 shows the 115-day Average RMSE for forecast of market value using RNN, LSTM, and GRU models and after reconciliation. The results demonstrate competitive performance across all methods. The minimum Average RMSE values for each level indicate that MinT (Wls-struct) consistently yields the best results, especially at the Total Aggregate and Industry levels. At the Company level, the Top-Down (Fp) method provides the lowest average RMSE for RNN while LSTM and GRU show similar performance overall, indicating that the choice of reconciliation method is crucial for achieving optimal accuracy in hierarchical forecasting. Before reconciliation, the average RMSE values for RNN, LSTM, and GRU models typically show higher variability across the different levels. However, after applying reconciliation methods, there is a consistent reduction in Average RMSE across most reconciliation methods. This reduction indicates that reconciliation effectively enhances the precision of forecasts in all levels. Despite some variations in performance across methods and models, the overall trend suggests that reconciliation contributes positively to reducing forecasting errors in the shorter 115-day forecasting horizon.

Table 5 shows the 230-day Average RMSE for forecast of market value using RNN, LSTM, and GRU models and after reconciliation. The results demonstrate

competitive performance across all methods. The minimum RMSE values for each level indicate that MinT (Wls-struct) consistently yields the best results for all levels.

Table 6 shows the 115-day Average MAPE for market value forecasts using RNN, LSTM, and GRU models with various reconciliation methods. The results indicate that the MinT (Wls-struct) method consistently yields the lowest MAPE across levels 0 and 1, while for level 2, the BU method showed superior performance.

Table 7 shows 230-day forecast Average MAPE values for forecasting market value using RNN, LSTM, and GRU models at different hierarchical levels with various reconciliation methods. Similar to the Average RMSE performance, MinT (Wls-struct) consistently achieves the lowest Average MAPE across all levels and models, reinforcing its effectiveness in hierarchical forecasting.

Table 8 shows AvgRelRMSE (115-day Forecast) for Market Value Using RNN, LSTM, and GRU with the reconciliation methods. A value of the AvgRelRMSE equal to one shows that the reconciled forecast is the same as the base forecast at some level. An example is the BU approach, which uses base forecasts from Level 2 and sums them to obtain forecasts for the higher levels. A value of AvgRelRMSE less than one shows that the reconciled forecasts showed improvements compared to the base forecasts. Bold values in Table 8 represent the lowest AvgRelRMSE, highlighting the best reconciliation method at a given level for a specific forecasting method. For RNN, LSTM, and GRU, improvements are seen with different reconciliation methods at different levels. Notably, the MT(Wls-struct) method consistently provides the most significant improvement in forecast accuracy across all models and levels, indicating its robustness and effectiveness in forecast reconciliation. For instance, for RNN, the AvgRelRMSE values are lowest at Level 0 (0.74990), Level 1 (0.74476), and Level 2 (0.93490) using the MT(Wls-struct) method. Similar trends are observed for LSTM and GRU, where MT(Wls-struct) achieves the best performance across all levels, such as 0.61171 at Level 1 for LSTM and 0.75735 at Level 0 for GRU. In summary, the MT(Wls-struct) method shows consistent and significant improvements over the base forecasts, demonstrating its superior performance in reconciling.

Table 4 115-day Forecast Average RMSE for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.28943	0.28705	0.28943	0.28838	0.25873	0.22854	0.25873
Level 1	0.09709	0.09160	0.08821	0.09709	0.08428	0.07279	0.08428
Level 2	0.03309	0.03309	0.03049	0.03298	0.03221	0.03075	0.03221
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.27539	0.27277	0.27539	0.38236	0.24288	0.22464	0.24288
Level 1	0.11091	0.08782	0.10248	0.11091	0.08759	0.07399	0.08759
Level 2	0.03210	0.03210	0.03372	0.03543	0.03210	0.03022	0.03210
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.31887	0.31827	0.31887	0.36727	0.27353	0.23547	0.27353
Level 1	0.10907	0.09535	0.09759	0.10907	0.08829	0.07393	0.08829
Level 2	0.03375	0.03375	0.03297	0.03579	0.03323	0.03116	0.03323

Table 5 230-day Forecast Average RMSE for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.32341	0.29866	0.32341	0.31219	0.28541	0.24253	0.28541
Level 1	0.10088	0.09143	0.09631	0.10088	0.08851	0.07513	0.08851
Level 2	0.03266	0.03266	0.03199	0.03354	0.03236	0.03053	0.03236
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.30112	0.30027	0.30112	0.49048	0.26691	0.25761	0.26691
Level 1	0.13190	0.09307	0.09810	0.13190	0.09710	0.08068	0.09710
Level 2	0.03330	0.03330	0.03315	0.04193	0.03421	0.03178	0.03421
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.31137	0.29947	0.31137	0.35575	0.27012	0.23581	0.27012
Level 1	0.10952	0.09207	0.09902	0.10952	0.09006	0.07621	0.09006
Level 2	0.03263	0.03263	0.03281	0.03673	0.03268	0.03078	0.03268

Table 6 115-day Forecast Average MAPE for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.65631	1.64351	1.65631	1.74494	1.50307	1.34749	1.50307
Level 1	2.26925	2.25791	2.60406	2.26925	4.86048	1.99688	4.86048
Level 2	3.24307	3.24307	3.66124	3.44672	6.57689	4.18384	6.57689
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.59992	1.58912	1.59992	2.23833	1.42022	1.30097	1.42022
Level 1	3.51583	2.17036	3.06996	3.51583	5.97304	2.14199	5.97304
Level 2	3.18023	3.18023	3.94552	4.38641	7.11633	4.37370	7.11633
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.98305	1.97369	1.98305	2.28538	1.72786	1.50005	1.72786
Level 1	2.78095	2.43671	2.79038	2.78095	6.31151	2.08197	6.31151
Level 2	3.35354	3.35354	3.70493	3.72860	7.40183	4.38271	7.40183

Table 9 shows AvgRelRMSE (230-day forecast) for market value using RNN, LSTM, and GRU with the reconciliation methods. Similar to the 115-day forecast, improvements are seen with different reconciliation methods at different levels, with MinT (Wls-struct) showing the most significant improvement in forecast accuracy across all models and levels, indicating its robustness and effectiveness in forecast reconciliation.

Table 10 displays the AveRelMAPE for market value using RNN, LSTM, and GRU models with various reconciliation methods across Levels 0, 1, and 2 for the 115-day forecast horizon. Lower AveRelMAPE values indicate better reconciliation and forecasting accuracy. For each prediction model, bold values in Table 10 highlight the lowest AveRelMAPE within the respective reconciliation methods, emphasizing the best performing method for each level. Similar to AvgRelRMSE, AveRelMAPE values close to 1 indicate that the reconciled forecast aligns closely with the base forecast, while values less than 1 suggest improvements due to the

Table 7 230-day Forecast Average MAPE for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	2.01405	1.87285	2.01405	1.98979	1.80411	1.54966	1.80411
Level 1	2.43746	2.42326	2.63387	2.43746	5.54618	2.13251	5.54618
Level 2	3.35565	3.35565	3.75758	3.53452	7.02998	4.37563	7.02998
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.89036	1.88575	1.89036	3.06671	1.67847	1.55793	1.67847
Level 1	3.56219	2.38515	2.97529	3.56219	6.90191	2.26663	6.90191
Level 2	3.34079	3.34079	3.82718	4.26944	7.85180	4.61958	7.85180
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.93981	1.87885	1.93981	2.21831	1.70245	1.49331	1.70245
Level 1	2.85120	2.38853	2.80420	2.85120	6.08199	2.13467	6.08199
Level 2	3.35735	3.35735	3.80820	3.65567	7.28949	4.37204	7.28949

Table 8 AvgRelRMSE (115-day Forecast) for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.92346	1.00000	0.96531	0.88251	0.74990	0.88251
Level 1	1.00000	0.90632	0.95472	1.00000	0.87732	0.74476	0.87732
Level 2	1.00000	1.00000	0.97940	1.02693	0.99073	0.93490	0.99073
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99718	1.00000	1.62886	0.88639	0.85550	0.88639
Level 1	1.00000	0.70564	0.74377	1.00000	0.73618	0.61171	0.73618
Level 2	1.00000	1.00000	0.99528	1.25913	1.02718	0.95429	1.02718
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.96180	1.00000	1.14256	0.86753	0.75735	0.86753
Level 1	1.00000	0.84060	0.90405	1.00000	0.82229	0.69584	0.82229
Level 2	1.00000	1.00000	1.00540	1.12548	1.00149	0.94314	1.00149

reconciliation methods. The results demonstrate notable improvements in forecast accuracy across different levels and models, particularly with the MinT (Wls-struct) method. This method consistently achieves the lowest AveRelMAPE across multiple levels, indicating its effectiveness in enhancing hierarchical time series forecasting accuracy compared to other reconciliation methods like BU, TD(Fp), and MO(Fp). The Table 11 presents AveRelMAPE values for RNN, LSTM, and GRU models using various reconciliation methods over a 230-day forecast. For each level (0, 1), while for level 2, the BU shows superior performance.

Tables 10 and 11 show the 1-AvgRelRMSE as a percentage for 115 and 230-day forecasts, respectively, indicating the relative change in forecast accuracy compared to a baseline, where higher values signify improved performance. Positive percentages reflect enhanced accuracy, while negative percentages indicate decreased accuracy compared to the baseline forecast.

Table 9 AvgRelRMSE (230-day forecast) for market value using RNN, LSTM, and GRU with reconciliation methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99178	1.00000	0.99640	0.89394	0.78964	0.89394
Level 1	1.00000	0.94346	0.90857	1.00000	0.86807	0.74973	0.86807
Level 2	1.00000	1.00000	0.92137	0.99676	0.97351	0.92923	0.97351
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99048	1.00000	1.38842	0.88195	0.81571	0.88195
Level 1	1.00000	0.79186	0.92405	1.00000	0.78977	0.66713	0.78977
Level 2	1.00000	1.00000	1.05058	1.10392	1.00002	0.94148	1.00002
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99812	1.00000	1.15181	0.85781	0.73845	0.85781
Level 1	1.00000	0.87423	0.89476	1.00000	0.80949	0.67777	0.80949
Level 2	1.00000	1.00000	0.97674	1.06029	0.98443	0.92314	0.98443

Table 10 AveRelMAPE (115-day forecast) for market value using RNN, LSTM, and GRU with reconciliation methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.92989	1.00000	0.98795	0.89576	0.76942	0.89576
Level 1	1.00000	0.99417	1.08058	1.00000	2.27539	0.87489	2.27539
Level 2	1.00000	1.00000	1.11978	1.05330	2.09497	1.30396	2.09497
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99756	1.00000	1.62229	0.88791	0.82414	0.88791
Level 1	1.00000	0.66957	0.83524	1.00000	1.93755	0.63630	1.93755
Level 2	1.00000	1.00000	1.14559	1.27797	2.35028	1.38278	2.35028
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.96858	1.00000	1.14357	0.87764	0.76982	0.87764
Level 1	1.00000	0.83773	0.98352	1.00000	2.13313	0.74869	2.13313
Level 2	1.00000	1.00000	1.13429	1.08885	2.17120	1.30223	2.17120

In the 115-day forecast, the MT(Wls-struct) method consistently demonstrates superior performance across all levels and models. At Level 0, MT(Wls-struct) improves accuracy by 25.01% for RNN, 14.45% for LSTM, and 24.27% for GRU. Similarly, at Level 1, the accuracy improvements are 25.52% for RNN, 38.83% for LSTM, and 30.42% for GRU. At Level 2, MT(Wls-struct) also shows positive results with 6.51% for RNN, 4.57% for LSTM, and 5.69% for GRU. On the other hand, the MO(Fp) method often leads to negative impacts on forecast accuracy, particularly for LSTM at Levels 0 and 2 with reductions of -62.89% and -25.91%, respectively, and for GRU at Levels 0 and 2 with reductions of -14.26% and -12.55%, respectively. This highlights MT(Wls-struct) as the most effective reconciliation method, while MO(Fp) frequently detracts from accuracy (Tables 12 and 13).

For the 230-day forecast, the MT(Wls-struct) method remains the top performer across all models and levels. At Level 0, MT(Wls-struct) achieves improvements of 21.04% for RNN, 18.43% for LSTM, and 26.16% for GRU. At Level 1, the improvements are 25.03% for RNN, 33.29% for LSTM, and 32.22% for GRU. At

Level 2, MT(Wls-struct) also shows positive results with 7.08% for RNN, 5.85% for LSTM, and 7.69% for GRU. The MO(Fp) method again results in negative impacts, particularly for LSTM at Levels 0 and 2 with reductions of -38.84% and -10.39%, respectively, and for GRU at Levels 0 and 2 with reductions of -15.18% and -6.03%, respectively. This further supports the effectiveness of MT(Wls-struct) and the frequent detriments of MO(Fp).

In conclusion, the MT(Wls-struct) method is consistently the most effective reconciliation approach across both 115-day and 230-day forecasts, showing the highest improvements in forecast accuracy for RNN, LSTM, and GRU models at all hierarchical levels. Conversely, the MO(Fp) method often leads to significant reductions in forecast accuracy, particularly for LSTM and GRU models. Therefore, MT(Wls-struct) is preferred for hierarchical reconciliation to achieve better forecast accuracy.

In evaluating the 1-AveRelMAPE percentages for both 115-day and 230-day forecasts across different reconciliation methods and models, several insights emerge. The 1-AveRelMAPE serves as a relative measure of forecast accuracy improvement or degradation compared to a baseline, where positive percentages denote enhancement and negative percentages indicate reduced accuracy.

Table 14 provides insights into the performance of RNN, LSTM, and GRU models with various reconciliation methods at different hierarchical levels for the 115-day forecast period. At Level 0, MT(Wls-struct) consistently demonstrates the highest improvement percentages across all models, indicating superior reconciliation effectiveness (RNN: 23.06%, LSTM: 17.59%, GRU: 23.02%). Conversely, methods like MO(Fp) show significant negative impacts on forecast accuracy, particularly pronounced in LSTM (-62.23% at Level 0) and GRU (-14.36% at Level 1).

Moving to Table 15, which focuses on the 230-day forecast period, a similar trend is observed. MT(Wls-struct) continues to outperform other methods, with notable improvements across all models and levels (e.g., LSTM: 18.685% at Level 0, 39.076% at Level 1).

These tables underscore the critical role of reconciliation methods in hierarchical forecasting accuracy. MT(Wls-struct) emerges as a robust choice, consistently enhancing forecast precision across different forecast lengths and model architectures.

Table 11 AveRelMAPE (230-day forecast) for market value using RNN, LSTM, and GRU with reconciliation methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (OLS), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99227	1.00000	1.05351	0.90748	0.81355	0.90748
Level 1	1.00000	0.99500	1.14754	1.00000	2.14189	0.87997	2.14189
Level 2	1.00000	1.00000	1.12894	1.06280	2.02798	1.29009	2.02798
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99325	1.00000	1.39903	0.88769	0.81315	0.88769
Level 1	1.00000	0.61731	0.87318	1.00000	1.69890	0.60924	1.69890
Level 2	1.00000	1.00000	1.24064	1.37928	2.23768	1.37528	2.23768
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	1.00000	0.99528	1.00000	1.15246	0.87131	0.75644	0.87131
Level 1	1.00000	0.87622	1.00339	1.00000	2.26955	0.74865	2.26955
Level 2	1.00000	1.00000	1.10478	1.11184	2.20717	1.30689	2.20717

Table 12 1-AVERELRMSE in % for 115-day forecast for market value using RNN, LSTM, and GRU with reconciliation methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.00%	7.65%	0.00%	3.47%	11.75%	25.01%	11.75%
Level 1	0.00%	9.37%	4.53%	0.00%	12.27%	25.52%	12.27%
Level 2	0.00%	0.00%	2.06%	-2.69%	0.93%	6.51%	0.93%
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.00%	0.28%	0.00%	-62.89%	11.36%	14.45%	11.36%
Level 1	0.00%	29.44%	25.62%	0.00%	26.38%	38.83%	26.38%
Level 2	0.00%	0.00%	0.47%	-25.91%	-2.72%	4.57%	-2.72%
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.00%	3.82%	0.00%	-14.26%	13.25%	24.27%	13.25%
Level 1	0.00%	15.94%	9.59%	0.00%	17.77%	30.42%	17.77%
Level 2	0.00%	0.00%	-0.54%	-12.55%	-0.15%	5.69%	-0.15%

Table 13 1-AvgRelRMSE in % for 230-day forecast for market value using RNN, LSTM, and GRU with reconciliation methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.000%	0.822%	0.000%	0.360%	10.606%	21.036%	10.606%
Level 1	0.000%	5.654%	9.143%	0.000%	13.193%	25.027%	13.193%
Level 2	0.000%	0.000%	7.863%	0.324%	2.649%	7.077%	2.649%
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.000%	0.952%	0.000%	-38.842%	11.805%	18.429%	11.805%
Level 1	0.000%	20.814%	7.595%	0.000%	21.023%	33.287%	21.023%
Level 2	0.000%	0.000%	-5.058%	-10.392%	-0.002%	5.852%	-0.002%
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.000%	0.188%	0.000%	-15.181%	14.219%	26.155%	14.219%
Level 1	0.000%	12.577%	10.524%	0.000%	19.051%	32.223%	19.051%
Level 2	0.000%	0.000%	2.326%	-6.029%	1.557%	7.686%	1.557%

5 Conclusion

Several key insights emerge from this study's analysis of hierarchical time series forecasting for market value, which employs RNN, LSTM, and GRU models, as well as various reconciliation methods. These insights address significant gaps in current research and practice.

The major goal of this study was to evaluate and compare the performance of RNN, LSTM, and GRU models at various hierarchical levels of market value forecasting using the DJIA market index. The levels include the total market value, industry-specific values, and individual stock market values. Reconciliation strategies tested include Bottom-Up (BU), Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols). Each method was evaluated using established forecast accuracy metrics, such as Average Root Mean Squared Error, Average Mean Absolute Percentage Error, Average Relative RMSE, and Average Relative MAPE, to determine their effectiveness in improving forecasting precision.

Table 14 1-AveRelMAPEin % for (115-day Forecast) for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.00%	7.01%	0.00%	1.20%	10.42%	23.06%	10.42%
Level 1	0.00%	0.58%	-8.06%	0.00%	-127.54%	12.51%	-127.54%
Level 2	0.00%	0.00%	-11.98%	-5.33%	-109.50%	-30.40%	-109.50%
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.00%	0.24%	0.00%	-62.23%	11.21%	17.59%	11.21%
Level 1	0.00%	33.04%	16.48%	0.00%	-93.75%	36.37%	-93.75%
Level 2	0.00%	0.00%	-14.56%	-27.80%	-135.03%	-38.28%	-135.03%
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.00%	3.14%	0.00%	-14.36%	12.24%	23.02%	12.24%
Level 1	0.00%	16.23%	1.65%	0.00%	-113.31%	25.13%	-113.31%
Level 2	0.00%	0.00%	-13.43%	-8.89%	-117.12%	-30.22%	-117.12%

Table 15 1-AveRelMAPE in % for (230-day Forecast) for Market Value Using RNN, LSTM, and GRU with Reconciliation Methods (Bottom-Up, Top-Down (Fp), Middle-Out (Fp), MinT (Ols), MinT (Wls-struct), and Optimal Combination (Ols)). The bold values show the best performing reconciliation method for each level

Level	RNN	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.000%	0.773%	0.000%	-5.351%	9.252%	18.645%	9.252%
Level 1	0.000%	0.500%	-14.754%	0.000%	-114.189%	12.003%	-114.189%
Level 2	0.000%	0.000%	-12.894%	-6.280%	-102.798%	-29.009%	-102.798%
Level	LSTM	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.000%	0.675%	0.000%	-39.903%	11.231%	18.685%	11.231%
Level 1	0.000%	38.269%	12.682%	0.000%	-69.890%	39.076%	-69.890%
Level 2	0.000%	0.000%	-24.064%	-37.928%	-123.768%	-37.528%	-123.768%
Level	GRU	BU	TD(Fp)	MO(Fp)	MT(Ols)	MT(Wls-struct)	OC(Ols)
Level 0	0.000%	0.472%	0.000%	-15.246%	12.869%	24.357%	12.869%
Level 1	0.000%	12.379%	-0.339%	0.000%	-126.955%	25.135%	-126.955%
Level 2	0.000%	0.000%	-10.478%	-11.184%	-120.717%	-30.689%	-120.717%

Throughout the tests, it became clear that reconciliation approaches play an important role in improving forecast accuracy in HTS applications. MinT (Wls-struct) is the most effective method for reducing Average RMSE and Average MAPE across different forecasting horizons, including 230 days (one trading year) and 115 days (half trading year). Furthermore, while comparing the performance of RNN, LSTM, and GRU models, each demonstrated significant strengths based on the reconciliation method used and the forecasting hierarchy. Despite setting all parameters identically for a fair comparison, RNN, LSTM, and GRU models performed similarly before and after reconciliation across different techniques. This discovery emphasizes their appropriateness for hierarchical forecasting assignments where complex hierarchical structures and changing market conditions need advanced modeling techniques.

The combination of reconciliation methodologies and the deep neural network reduces forecast variability while simultaneously increasing overall prediction reliability across a wide range of forecasting scenarios. By systematically evaluating

these methodologies over both short-term (115 days) and long-term (230 days), this study provides practitioners with actionable insights into selecting optimal forecasting strategies tailored to specific hierarchical forecasting contexts, thereby filling a critical research gap in the application of deep learning methods to hierarchical time series forecasting. These insights not only contribute to advancing theoretical knowledge but also offer practical guidance for practitioners seeking to leverage cutting-edge methodologies for enhanced decision-making in financial and business forecasting domains.

6 Limitations and Recommendations

This study has several limitations. Firstly, the analysis was conducted using a single dataset, the DJIA market index, which may restrict the generalizability of the findings to other market indices or financial contexts. The unique characteristics of the DJIA might not fully represent the behavior of other markets or sectors, potentially limiting the applicability of the results. Additionally, the process of training and testing models across a large number of hierarchical series is time-consuming and computationally intensive. This aspect could be a constraint when applying these methods to larger or more complex datasets. Parameter optimization for the models, while crucial for achieving accurate forecasts, also adds another layer of complexity and resource demand.

To address these limitations and enhance the robustness of future research, it is recommended to explore hybrid models that integrate the strengths of different forecasting approaches. Combining neural network architectures with advanced reconciliation methods could potentially improve forecast accuracy and adaptability. Furthermore, Transformer models, which have shown promising results in time series forecasting tasks, could be investigated as part of future work. These models may provide an alternative to RNN, LSTM, and GRU models, potentially offering better performance due to their ability to capture long-range dependencies and parallel processing capabilities.

Another recommendation is to conduct ablation studies that evaluate the impact of different components, such as the choice of reconciliation method, model architecture, and preprocessing techniques, on forecast performance. This would provide a clearer understanding of which elements contribute most to the accuracy of the predictions.

Moreover, the application of clustering methods for grouping similar time series data prior to model training could help reduce noise and improve forecasting performance, especially in hierarchical settings. Future studies should consider using diverse datasets across various market indices to validate and compare the findings. This will help in assessing the generalizability of the models and reconciliation techniques. Additionally, comparative analyses involving different forecasting methods and hierarchical structures will further contribute to the refinement of forecasting strategies and provide a more comprehensive understanding of their effectiveness in varied contexts.

7 Abbreviations

The following abbreviations are used in this manuscript:

HTS	Hierarchical Time Series
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
BU	Bottom-Up
TD(Fp)	Top-Down (Forecast Proportions)
MO(Fp)	Middle-Out (Forecast Proportions)
MT(Ols)	Minimum Trace (Ordinary Least Squares)
MT(Wls-struct)	Minimum Trace (Weighted Least Squares Structural)
OC(Ols)	Optimal Combination (Ordinary Least Squares)
DJIA	Dow Jones Industrial Average
MAPE	Mean Absolute Percentage Error
RMSE	Root Mean Squared Error
AvgRelRMSE	Average Relative Root Mean Squared Error
AveRelMAPE	Average Relative Mean Absolute Percentage Error

Author Contributions **John Kamwele Mutinda:** Writing-Original draft, Writing-Review, Editing, Software, Methodology, Data curation and Conceptualization. **Li Yong:** Writing-Review and Supervision.

Funding This research received no external funding.

Data Availability The data and Implementation materials are available upon request.

Declarations

Consent for Publication Not applicable.

References

- Abolghasemi, M., Hyndman, R. J., Spiliotis, E., & Bergmeir, C. (2022a). Model selection in reconciling hierarchical time series. *Machine Learning*, 1–51.
- Abolghasemi, M., Tarr, G., & Bergmeir, C. (2022b). Machine learning applications in hierarchical time series forecasting: Investigating the impact of promotions. *International Journal of Forecasting*.
- Ahmadi, A., Nabipour, M., Mohammadi-Ivatloo, B., Amani, A. M., Rho, S., & Piran, M. J. (2020). Long-term wind power forecasting using tree-based learning algorithms. *IEEE Access*, 8, 151511–151522.
- Anderson, T. W. (2011). *The statistical analysis of time series*. John Wiley & Sons.
- Asif, S., Khan, S. U. R., Amjad, K., & Awais, M. (2024). Skinc-net: An efficient lightweight deep learning model for multiclass skin lesion classification in dermoscopic images. *Multimedia Tools and Applications*, 1–27.
- Athanasopoulos, G., Ahmed, R. A., & Hyndman, R. J. (2009). Hierarchical forecasts for Australian domestic tourism. *International Journal of Forecasting*, 25(1), 146–166.
- Bala, R., Singh, R. P., et al. (2019). Financial and non-stationary time series forecasting using lstm recurrent neural network for short and long horizon. In *2019 10th international conference on computing, communication and networking technologies (ICCCNT)* (pp. 1–7). IEEE.
- Bontempi, G. (2013). Machine learning strategies for time series prediction. *European Business Intelligence Summer School, Hammamet, Lecture* (pp. 1–128).

- Box, G. E., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time series analysis: Forecasting and control*. John Wiley & Sons.
- Bullinaria, J. A. (2013). Recurrent neural networks. *Neural Computation: Lecture*, 12(1).
- Chan, N. H. (2004). *Time series: Applications to finance*. John Wiley & Sons.
- Chen, Z., Ma, Q., & Lin, Z. (2021). Time-aware multi-scale rnns for time series modeling. In *IJCAI* (pp. 2285–2291).
- Chengzhao, Z., Heping, P., Yu, M., & Xun, H. (2019). Analysis of Asia Pacific stock markets with a novel multiscale model. *Physica A: Statistical Mechanics and its Applications*, 534, 120939.
- Cochrane, J. H. (1997). *Time series for macroeconomics and finance*.
- Dangerfield, B. J., & Morris, J. S. (1992). Top-down or bottom-up: Aggregate versus disaggregate extrapolations. *International Journal of Forecasting*, 8(2), 233–241.
- Davydenko, A., & Fildes, R. (2013). Measuring forecasting accuracy: The case of judgmental adjustments to sku-level demand forecasts. *International Journal of Forecasting*, 29(3), 510–522.
- Deng, T., He, X., & Zeng, Z. (2018). Recurrent neural network for combined economic and emission dispatch. *Applied Intelligence*, 48, 2180–2198.
- Dey, R., & Salem, F. M. (2017). Gate-variants of gated recurrent unit (gru) neural networks. In *2017 IEEE 60th international midwest symposium on circuits and systems (MWSCAS)* (pp. 1597–1600). IEEE.
- Fliedner, G. (1999). An investigation of aggregate variable time series forecast strategies with specific subaggregate time series statistical correlation. *Computers & Operations Research*, 26(10–11), 1133–1149.
- Gleason, J. L. (2020). Forecasting hierarchical time series with a regularized embedding space. *San Diego*, 7, 6–2.
- Gross, C. W., & Sohl, J. E. (1990). Disaggregation methods to expedite product line forecasting. *Journal of Forecasting*, 9(3), 233–254.
- Hochreiter, S., & Schmidhuber, J. (1997a). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>. ISSN 0899-7667
- Hochreiter, S., & Schmidhuber, J. (1997b). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and practice*. OTexts.
- Hyndman, R. J., Ahmed, R. A., Athanasopoulos, G., & Shang, H. L. (2011). Optimal combination forecasts for hierarchical time series. *Computational Statistics & Data Analysis*, 55(9), 2579–2589.
- Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for r. *Journal of Statistical Software*, 27, 1–22.
- Karevan, Z., & Suykens, J. A. (2020). Transductive lstm for time-series prediction: An application to weather forecasting. *Neural Networks*, 125, 1–9.
- Karmy, J. P., & Maldonado, S. (2019). Hierarchical time series forecasting via support vector regression in the European travel retail industry. *Expert Systems with Applications*, 137, 59–73.
- Khan, S. U. R., Asif, S., Bilal, O., & Ali, S. (2024). Deep hybrid model for mpox disease diagnosis from skin lesion images. *International Journal of Imaging Systems and Technology*, 34(2), Article e23044.
- Kinney, W. R. (1971). Predicting earnings: Entity versus subentity data. *Journal of Accounting Research* (pp. 127–136).
- Lipton, Z. C., Berkowitz, J., & Elkan, C. (2015). A critical review of recurrent neural networks for sequence learning. [arXiv:1506.00019](https://arxiv.org/abs/1506.00019).
- Liu, Z., Jiang, P., Zhang, L., & Niu, X. (2020). A combined forecasting model for time series: Application to short-term wind speed forecasting. *Applied Energy*, 259, Article 114137.
- Loureiro, A. L., Miguéis, V. L., & Da Silva, L. F. (2018). Exploring the use of deep neural networks for sales forecasting in fashion retail. *Decision Support Systems*, 114, 81–93.
- Mahkya, D., Ulama, B., et al. (2017). Hierarchical time series bottom-up approach for forecast the export value in central java. In *Journal of Physics: Conference Series* (vol. 893, pp. 012033). IOP Publishing.
- Mancuso, P., Piccialli, V., & Sudoso, A. M. (2021). A machine learning approach for forecasting hierarchical time series. *Expert Systems with Applications*, 182, Article 115102.
- Mandic, D. P., & Chambers, J. A. (1999). Exploiting inherent relationships in rnn architectures. *Neural Networks*, 12(10), 1341–1345.
- Mircetic, D., Rostami-Tabar, B., Nikolicic, S., & Maslaric, M. (2022). Forecasting hierarchical time series in supply chains: An empirical investigation. *International Journal of Production Research*, 60(8), 2514–2533.
- Mudelsee, M. (2010). Climate time series analysis. *Atmospheric and*, 397.

- Mudelsee, M. (2019). Trend analysis of climate time series: A review of methods. *Earth-science Reviews*, 190, 310–322.
- Nenova, Z. D., & May, J. H. (2016). Determining an optimal hierarchical forecasting model based on the characteristics of the data set. *Journal of Operations Management*, 44, 62–68.
- Oliveres, K. G., Garza, F., Luo, D., Challú, C., Mergenthaler, M., Taieb, S. B., Wickramasuriya, S. L., & Dubrawski, A. (2022). HierarchicalForecast: A reference framework for hierarchical forecasting in python. *Work in progress paper, submitted to Journal of Machine Learning Research*. [arXiv:2207.03517](https://arxiv.org/abs/2207.03517)
- Peng, Z., Peng, S., Fu, L., Lu, B., Tang, J., Wang, K., & Li, W. (2020). A novel deep learning ensemble model with data denoising for short-term wind speed forecasting. *Energy Conversion and Management*, 207, Article 112524.
- Pirani, M., Thakkar, P., Jivrani, P., Bohara, M. H., & Garg, D. (2022). A comparative analysis of arima, gru, lstm and bilstm on financial time series forecasting. In *2022 IEEE International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE)* (pp. 1–6). IEEE.
- Shahzad, I., Khan, S. U. R., Waseem, A., Abideen, Z. U., & Liu, J. (2024). Enhancing asd classification through hybrid attention-based learning of facial features. *Signal, Image and Video Processing*, 1–14.
- Shang, H. L., & Hyndman, R. J. (2017). Grouped functional time series forecasting: An application to age-specific mortality rates. *Journal of Computational and Graphical Statistics*, 26(2), 330–343.
- Shen, G., Tan, Q., Zhang, H., Zeng, P., & Xu, J. (2018). Deep learning with gated recurrent unit networks for financial sequence predictions. *Procedia Computer Science*, 131, 895–903.
- Silveira Gontijo, T., & Azevedo Costa, M. (2020). Forecasting hierarchical time series in power generation. *Energies*, 13(14), 3722.
- Spiliotis, E., Abolghasemi, M., Hyndman, R. J., Petropoulos, F., & Assimakopoulos, V. (2021). Hierarchical forecast reconciliation with machine learning. *Applied Soft Computing*, 112, Article 107756.
- Spiliotis, E., Petropoulos, F., & Assimakopoulos, V. (2019). Improving the forecasting performance of temporal hierarchies. *PLoS One*, 14(10), e0223422.
- Thomson, D. J. (1990). Time series analysis of holocene climate data. *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 330(1615), 601–616.
- Tsay, R. S. (2005). *Analysis of financial time series*. John Wiley & Sons.
- Villegas, M. A., & Pedregal, D. J. (2018). Supply chain decision support systems based on a novel hierarchical forecasting approach. *Decision Support Systems*, 114, 29–36.
- Wang, K., Qi, X., & Liu, H. (2019). Photovoltaic power forecasting based lstm-convolutional network. *Energy*, 189, Article 116225.
- Wan, H., Guo, S., Yin, K., Liang, X., & Lin, Y. (2020). Cts-lstm: Lstm-based neural networks for correlated time series prediction. *Knowledge-Based Systems*, 191, 105239.
- Wang, J., & Wang, J. (2015). Forecasting stock market indexes using principle component analysis and stochastic time effective neural networks. *Neurocomputing*, 156, 68–78.
- Wickramasuriya, S. L., Athanasopoulos, G., & Hyndman, R. J. (2018). Optimal forecast reconciliation for hierarchical and grouped time series through trace minimization. *Journal of the American Statistical Association*.
- Widiarta, H., Viswanathan, S., & Piplani, R. (2009). Forecasting aggregate demand: An analytical evaluation of top-down versus bottom-up forecasting in a production planning framework. *International Journal of Production Economics*, 118(1), 87–94.
- Wu, W., Liao, W., Miao, J., & Du, G. (2019). Using gated recurrent unit network to forecast short-term load considering impact of electricity price. *Energy Procedia*, 158, 3369–3374.
- Xu, Y., & Cohen, S. B. (2018). Stock movement prediction from tweets and historical prices. In *Proceedings of the 56th annual meeting of the association for computational linguistics (Volume 1: Long Papers)* (pp. 1970–1979).
- Yagli, G. M., Yang, D., & Srinivasan, D. (2019). Reconciling solar forecasts: Sequential reconciliation. *Solar Energy*, 179, 391–397.
- Yamak, P. T., Yujian, L., & Gadosey, P. K. (2019). A comparison between arima, lstm, and gru for time series forecasting. In *Proceedings of the 2019 2nd international conference on algorithms, computing and artificial intelligence* (pp. 49–55).
- Ye, L., Zhang, C., Tang, Y., Zhong, W., Zhao, Y., Lu, P., Zhai, B., Lan, H., & Li, Z. (2019). Hierarchical model predictive control strategy based on dynamic active power dispatch for wind power cluster integration. *IEEE Transactions on power systems*, 34(6), 4617–4629.
- Yu, Y., Si, X., Hu, C., & Zhang, J. (2019). A review of recurrent neural networks: Lstm cells and network architectures. *Neural Computation*, 31(7), 1235–1270.

- Zheng, Z., Chen, H., & Luo, X. (2019). Spatial granularity analysis on electricity consumption prediction using lstm recurrent neural network. *Energy Procedia*, 158, 2713–2718.
- Zheng, T., Salganik, M. J., & Gelman, A. (2006). How many people do you know in prison? Using over-dispersion in count data to estimate social structure in networks. *Journal of the American Statistical Association*, 101(474), 409–423.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.